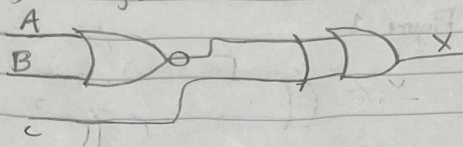


2b.) A three-input XNOR Gate

Logic Diagram



Truth Table

A	B	C	$(A \cdot B) \oplus C$
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

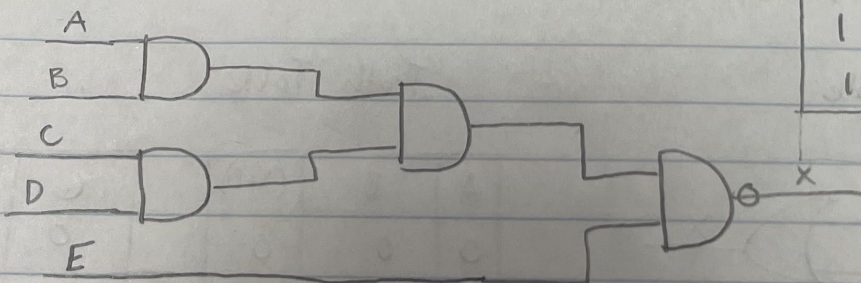
Boolean Expression

$$(A+B)' \oplus C$$

NAND Truth Table

A	B	C	D	E	output
0	0	0	0	0	1
0	0	0	0	1	1
0	0	0	1	0	1
0	0	1	0	0	1
0	1	0	0	0	1
1	0	0	0	0	1
0	0	0	1	1	1
0	0	1	1	0	1
0	1	1	0	0	1
1	1	0	0	0	1
0	0	1	1	1	1
0	1	1	1	0	1
1	1	1	0	0	1
0	1	1	1	1	1
1	1	1	1	0	1
1	1	1	1	1	0

2c.) A Five-input NAND Gate



Logic Diagram

Boolean Expression

$$X = (A \cdot B \cdot C \cdot D \cdot E)'$$